SQL QUERY GENERATION USING STRUCTURED PROMPTING

Aim

To design a structured prompt containing database schema, table details, and query requirements, then generate and validate an appropriate SQL query.

Procedure

1. Open Google Colab and create a notebook.

2. Provide the schema, table names, and column names in the prompt.

3. State the required output columns, filter condition, and sorting order.

4. Ask an LLM to generate SQL and review it for valid schema references.

5. Run the demonstration code to display the query.

6. Check that the query contains the requested table, filter, and ordering. In a real database, execute it against the matching schema.

Structured Prompt

```
Role: You are an SQL expert.
Schema:
- Students(student_id, student_name, department, cgpa)
- Enrollments(enrollment_id, student_id, course_id, marks)

Requirement: Return each student's name, department, and CGPA
for students whose CGPA is at least 8.0. Sort highest CGPA first.

Constraints:
- Use standard SQL.
- Use only listed tables and columns.
- Output the SQL query and a brief explanation.
- Do not invent schema elements.
- Check the filter and sorting against the requirement.
```

Program / Demonstration

```
query = """
SELECT s.student_name, s.department, s.cgpa
FROM Students AS s
WHERE s.cgpa >= 8.0
ORDER BY s.cgpa DESC;
"""

print("Schema: Students(student_id, student_name, department, cgpa)")
print("Requirement: CGPA >= 8.0, highest first")
print("\\nGenerated SQL query:")
print(query.strip())

# Basic validation of query requirements
assert "SELECT" in query.upper()
assert "FROM STUDENTS" in query.upper()
assert "CGPA >= 8.0" in query.upper()
assert "ORDER BY S.CGPA DESC" in query.upper()
print("\\nValidation passed: table, filter, and sorting are present.")
```

Google Colab Output Screenshot

Illustrative screenshot only, not captured from a live Colab session. Run the code in Colab for an authentic screenshot if required.

Google Colab SQL_Query_Generation.ipynb

[1]
```
query = '''
SELECT s.student_name, s.department, s.cgpa
FROM Students AS s
WHERE s.cgpa >= 8.0
ORDER BY s.cgpa DESC;
'''
print(query.strip())
```

Output

```
Generated SQL query:
SELECT s.student_name, s.department, s.cgpa
FROM Students AS s
WHERE s.cgpa >= 8.0
ORDER BY s.cgpa DESC;
Validation passed: requirements present.
```

Illustrative Colab-style screenshot; run the code for a live screenshot.

Expected Output

```
Generated SQL query:
SELECT s.student_name, s.department, s.cgpa
FROM Students AS s
WHERE s.cgpa >= 8.0
ORDER BY s.cgpa DESC;

Validation passed: table, filter, and sorting are present.
```

Result

The structured prompt included the schema and exact query requirements. The sample SQL selects the required student details, filters CGPA values of 8.0 or higher, and sorts them from highest to lowest. Text checks passed; execution against a real database is needed for full validation.